Algebraic Methods

Starter

Solution 2

1. Factorise each polynomial.

(a) $x^2 - 6x + 5$

1a $(x-1)(x-5)$

(b) $x^2 - 16$

1b $(x+4)(x-4)$

(c) $9x^2 - 25$

1c $(3x+5)(3x-5)$

2. Simplify fully.

(a) $\dfrac{x^2-9}{x^2+9x+18}$

2a $\dfrac{x-3}{x+6}$

(b) $\dfrac{2x^2+5x-12}{6x^2-7x-3}$

2b $\dfrac{x+4}{3x+1}$

(c) $\dfrac{x^2-x-30}{-x^2+3x+18}$

2c $-\dfrac{x+5}{x+3}$

3. Odd, even, or either?

For any integers n, m, decide whether each expression is always odd, always even, or could be either.

(a) $8n$

3a Always even.

(b) $n - m$

3b Either.

(c) $3m$

3c Either.

(d) $2n - 5$

3d Always odd.

Chapter Overview

Apart from the initial portion on proof, this chapter concerns manipulation of algebraic fractions.

1: Proof By Contradiction

Prove by contradiction that $\sqrt{5}$ is an irrational number.

2: $+ \div \times -$ Algebraic Fractions

Express

$$\frac{3x+5}{x^2+x-12} - \frac{2}{x-3}$$

as a single fraction in its simplest form.

3: Express a fraction using partial fractions.

Express $\frac{2+3x}{(x+3)(x-1)}$ in the form $\frac{A}{x+3} + \frac{B}{x-1}$ where A and B are constants.

4: Divide algebraic expressions, and convert an improper fraction into partial fraction form.

Given that

$$\frac{x^3+x^2+7}{x-3} \equiv Ax^2 + Bx + C + \frac{D}{x-3}$$

find the values of A, B, C and D.

1: Proof By Contradiction

Prove that there is no greatest odd integer.

Assumption Assume that there is a greatest odd integer, n.

Show contradiction

Then n + 2 is an odd integer larger than n.

This contradicts the assumption that n is the greatest odd integer.

Conclusion

Therefore, there is no greatest odd integer.

How to structure/word proof

"Assume that [negation of statement]."

[Reasoning followed by...] "This contradicts the assumption that..." or "This is a contradiction."

"Therefore [restate original statement]."

Proof by contradiction

Assume the statement is false. Prove this leads to a contradiction. The original statement must therefore be true.

Negating the original statement

Solution 5

The first part of a proof by contradiction requires you to negate the original statement. What is the negation of each statement? Click to choose.

There are infinitely many prime numbers.	**All Popes are Catholic.**	**If it is raining, my garden is wet.**
There are infinitely many non-prime (i.e. composite) numbers.	There exists a Pope who is not Catholic.	It is not raining and my garden is dry.
There are finitely many prime numbers.	No Popes are Catholic.	It is not raining and my garden is wet.
There are finitely many non-composite numbers.	Your Teacher is the Pope.	It is raining and my garden is not wet.

The negation of "all are" is "not all are", not "none are". For example, "not everyone likes green". Do not confuse negation with an opposite.

The negation of "if A then B" is "A and not B": the condition is true but the conclusion is false.

More Examples

Solution 6

Prove by contradiction that if n^2 is even, then n must be even.

Assumption

Assume there exists a number n such that n^2 is even, but n is not even.

Show contradiction

n is odd, therefore $n = 2k + 1$ for some integer k.

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1$$
$$= 2(2k^2 + 2k) + 1,$$

which is odd.

This contradicts the assumption that n^2 is even.

Conclusion

Therefore if n^2 is even then n must be even.

Negating a conditional statement

The negation of "if A then B" is "A and not B".

More Examples

Solution 7

Prove by contradiction that $\sqrt{2}$ is an irrational number.

Assumption

Assume that $\sqrt{2}$ is a rational number.

Show contradiction

Then $\sqrt{2} = \dfrac{a}{b}$, where a, b are integers, $b \neq 0$, and the fraction is in its simplest form.

$$2 = \frac{a^2}{b^2} \quad \Rightarrow \quad 2b^2 = a^2.$$

Thus a^2 is even, so a is even. Write $a = 2k$, where k is an integer.

$$2b^2 = (2k)^2 = 4k^2 \quad \Rightarrow \quad b^2 = 2k^2.$$

Thus b^2 is even, so b is even. Both a and b are even, contradicting that a/b is in its simplest form.

Conclusion

Therefore $\sqrt{2}$ is irrational.

Rational and irrational numbers

A rational number can be expressed as a/b, with integers a, b and $b \neq 0$. An irrational number cannot: examples include $\pi, e, \sqrt{3}$.

Rational numbers: $\mathbb{Q}$; real numbers: $\mathbb{R}$; natural numbers: $\mathbb{N}$; integers: $\mathbb{Z}$.

Alternative proof

In $2b^2 = a^2$, the power of 2 in the prime factorisation is odd on the left and even on the right. This is a contradiction.

More Examples

Solution 8

Prove by contradiction that there are infinitely many prime numbers.

Assumption

Assume that there is a finite number of prime numbers.

Show contradiction

List all the primes as $p_1, p_2, \ldots, p_n$. Consider

$N = p_1 p_2 \cdots p_n + 1$.

Division of N by any listed prime leaves remainder 1, so none of them divides N.

Either N is prime itself, or its prime factorisation contains primes not in the list. This contradicts the assumption that the list contains all primes.

Conclusion

Therefore, there are an infinite number of primes.

Euclid's proof

This is one of the earliest known proofs.

Each stage can be revealed separately: assumption, contradiction, then conclusion.

Exercise 5 Page 4 — Solution 9

5(a) Prove that if ab is irrational, then at least one of a, b is irrational. (3 marks)

5a

Assume ab is irrational but both a, b are rational. Write

$$a = \frac{c}{d}, \qquad b = \frac{e}{f},$$

where c, d, e, f are integers and $d, f \neq 0$. Then

$$ab = \frac{ce}{df}$$

is rational, since ce, df are integers and $df \neq 0$. This contradicts the assumption.

Therefore at least one of a, b is irrational.

5(b) Prove that if $a + b$ is irrational, then at least one of a, b is irrational. (3 marks)

5b

Assume $a + b$ is irrational but both a, b are rational. Write

$$a = \frac{c}{d}, \qquad b = \frac{e}{f},$$

where c, d, e, f are integers and $d, f \neq 0$. Then

$$a + b = \frac{cf + de}{df}$$

is rational. This contradicts the assumption.

Therefore at least one of a, b is irrational.

Exercise 6 Page 4 — Solution 10

6. Use proof by contradiction to show that there exist no integers a, b for which $21a + 14b = 1$.

Working

Assume there exist integers a, b such that $21a + 14b = 1$. Dividing by 7 gives

$$3a + 2b = \frac{1}{7}.$$

Since $3a$ and $2b$ are integers, their sum is an integer. But $1/7$ is not an integer: a contradiction.

Therefore there are no integers a, b for which $21a + 14b = 1$.

Exercise 7a Page 4 — Solution 11

7(a) Prove by contradiction that if n^2 is a multiple of 3, n is a multiple of 3. (3 marks)

7(b) Hence prove by contradiction that $\sqrt{3}$ is irrational. (3 marks)

Working

Part (a) Assume n^2 is a multiple of 3 but n is not. Then $n = 3k + 1$ or $n = 3k + 2$ for an integer k.

$(3k + 1)^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1,$

$(3k + 2)^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1.$

In either case the square has remainder 1 on division by 3. This contradicts the assumption that n^2 is a multiple of 3.

Therefore, if n^2 is a multiple of 3, so is n.

Exercise 7b Page 4 — Solution 12

7(a) Prove by contradiction that if n^2 is a multiple of 3, n is a multiple of 3. (3 marks)

7(b) Hence prove by contradiction that $\sqrt{3}$ is irrational. (3 marks)

Working

Part (b) Assume $\sqrt{3} = a/b$ for integers a, b, with $b \neq 0$, in simplest terms. Then

$3 = \dfrac{a^2}{b^2} \quad \Rightarrow \quad a^2 = 3b^2.$

By part (a), a is a multiple of 3. Write $a = 3c$, where c is an integer.

$9c^2 = 3b^2 \quad \Rightarrow \quad b^2 = 3c^2.$

By part (a), b is also a multiple of 3. Thus a, b have a common factor of 3, contradicting the fraction being in simplest terms.

Therefore $\sqrt{3}$ is irrational.

Exercise 1A Page 4-5

Solution 13

8. Use proof by contradiction to prove: "There are no integer solutions to $x^2 - y^2 = 2$."

Working

Assume there is an integer solution. Then $(x-y)(x+y) = 2$. The integer factor pairs of 2 give:

$x-y$	$x+y$	x	y
2	1	$3/2$	$-1/2$
1	2	$3/2$	$1/2$
-2	-1	$-3/2$	$1/2$
-1	-2	$-3/2$	$-1/2$

None gives integer values of x, y, contradicting the assumption.

Therefore there are no integer solutions to $x^2 - y^2 = 2$.

Exercise 9 Page 4

Solution 14

9. Prove by contradiction that $\sqrt[3]{2}$ is irrational.

Working

Assume $\sqrt[3]{2} = a/b$, where a, b are integers, $b \neq 0$, and the fraction is in simplest terms. Then $a^3 = 2b^3$, so a^3 is even and hence a is even. Write $a = 2n$, where n is an integer.

$$(2n)^3 = 2b^3,$$
$$8n^3 = 2b^3,$$
$$b^3 = 4n^3.$$

Thus b^3 is even, so b is even too. This gives a common factor of 2, contradicting that a/b is in simplest terms.

Therefore $\sqrt[3]{2}$ is irrational.

Pearson Pure Mathematics Year 2/AS Pages 4-5 — Solution 15

Extension

[STEP I 2008 Q1] What does it mean to say that a number x is rational?

Prove by contradiction statements A and B below, where p and q are real numbers.

A: If pq is irrational, then at least one of p and q is irrational.

B: If $p+q$ is irrational, then at least one of p and q is irrational.

Disprove by means of a counterexample statement C below, where p and q are real numbers.

C: If p and q are irrational, then $p+q$ is irrational.

If the numbers e, π, π^2, e^2 and $e\pi$ are irrational, prove that at most one of the numbers $\pi+e, \pi-e, \pi^2-e^2, \pi^2+e^2$ is rational.

Solutions on next slide.

Note: e is Euler's Number and you will encounter it later at A Level. But for the moment, you only need to know it is an irrational number, like π.

Solution to Extension Problem — Solution 16

A number x is rational if $x = m/n$ for integers m, n with $n \neq 0$. An irrational number cannot be written this way.

A. If pq is irrational, at least one of p, q is irrational.

A

Assume pq is irrational but p, q are both rational. Their product is then rational, contradicting the assumption. Therefore A is true.

B. If $p+q$ is irrational, at least one of p, q is irrational.

B

Assume $p+q$ is irrational but p, q are both rational. Their sum is then rational, contradicting the assumption. Therefore B is true.

C. Disprove: if p, q are irrational, then $p+q$ is irrational.

C

Take $p = \sqrt{2}$ and $q = -\sqrt{2}$. Both are irrational, but $p+q = 0$ is rational. This is a counterexample.

Solution to Extension Problem

Solution 17

Assuming $e, \pi, \pi^2, e^2, e\pi$ are irrational, prove that at most one of $\pi + e, \pi - e, \pi^2 - e^2, \pi^2 + e^2$ is rational.

First rule out pairs

$(\pi + e) + (\pi - e) = 2\pi$ is irrational, so $\pi + e$ and $\pi - e$ cannot both be rational.

$(\pi^2 + e^2) + (\pi^2 - e^2) = 2\pi^2$ is irrational, so these two cannot both be rational.

If $\pi + e$ and $\pi^2 - e^2$ were rational, then

$$\pi - e = \frac{\pi^2 - e^2}{\pi + e}$$

would be rational, a contradiction. Similarly, $\pi - e$ and $\pi^2 - e^2$ cannot both be rational.

The remaining possible pairs

If two were rational, they would have to be $\pi^2 + e^2$ and one of $\pi \pm e$.

If $\pi + e$ and $\pi^2 + e^2$ were rational, then

$$2e\pi = (\pi + e)^2 - (\pi^2 + e^2)$$

would be rational, contradicting the assumption.

If $\pi - e$ and $\pi^2 + e^2$ were rational, then

$$2e\pi = (\pi^2 + e^2) - (\pi - e)^2$$

would be rational, again a contradiction.

Therefore at most one of the four numbers is rational.

2a: Multiplying/Dividing Algebraic Fractions

Solution 18

Multiply algebraic fractions as you would numerical fractions. Factorise and cancel common factors where possible.

$$\frac{a}{b} \times \frac{c}{a} = \frac{ac}{ab} = \frac{c}{b}$$

$$\frac{x+1}{2} \times \frac{3}{x^2-1} = \frac{3(x+1)}{2(x+1)(x-1)} = \frac{3}{2(x-1)}$$

To divide by a fraction, multiply by the reciprocal of the second fraction.

$$\frac{p}{q} \div \frac{r}{q} = \frac{p}{q} \times \frac{q}{r} = \frac{p}{r}$$

$$\frac{x+2}{x+4} \div \frac{3x+6}{x^2-16} = \frac{x+2}{x+4} \times \frac{(x+4)(x-4)}{3(x+2)} = \frac{x-4}{3}$$

Test Your Understanding — Solution 19

$$\frac{x+3}{5} \times \frac{10}{x^2-9} =$$

$$\frac{x+3}{5} \times \frac{10}{(x+3)(x-3)} = \frac{2}{x-3}$$

$$\frac{x^2+x}{y} \div \frac{x^2-x-2}{y^2} =$$

$$\frac{x(x+1)}{y} \times \frac{y^2}{(x+1)(x-2)} = \frac{xy}{x-2}$$

Spot the error

A student writes $\dfrac{x^2+y}{2y} = \dfrac{x^2}{2}$. Explain why this is incorrect.

Explanation

Cancellation means dividing numerator and denominator by a common factor. The student has subtracted y from the numerator. In general,

$$\frac{x^2+y}{2y} = \frac{x^2}{2y} + \frac{1}{2}, \qquad y \neq 0.$$

Exercise 1B (page 6-7) — Solution 20

3. Show that

$$\frac{x^2-64}{x^2-36} \div \frac{64-x^2}{x^2-36} = -1.$$

Working

For the original expression to be defined, $x \neq \pm 6, \pm 8$. Multiply by the reciprocal:

$$\frac{x^2-64}{x^2-36} \div \frac{64-x^2}{x^2-36} = \frac{x^2-64}{x^2-36} \times \frac{x^2-36}{64-x^2}$$

$$= \frac{x^2-64}{-(x^2-64)} = -1.$$

Exercise 1B (page 6-7) Solution 21

4. Show that

$$\frac{2x^2-11x-40}{x^2-4x-32}\times\frac{x^2+8x+16}{6x^2-3x-45}\div\frac{8x^2+20x-48}{10x^2-45x+45}=\frac{a}{b},$$

and find the integer constants a, b.

Working

Factorise each polynomial and multiply by the reciprocal:

$$\frac{(2x+5)(x-8)}{(x-8)(x+4)}\times\frac{(x+4)^2}{3(2x+5)(x-3)}\times\frac{5(2x-3)(x-3)}{4(2x-3)(x+4)}$$

$$=\frac{5}{12}.$$

Thus $a=5,\ b=12$ in lowest terms.

The original restrictions are $x\neq 8,-4,-\frac{5}{2},3,\frac{3}{2}$.

2b: Adding/Subtracting Algebraic Fractions Solution 22

To add or subtract fractions, find a common denominator.

Simplify $\frac{3}{x+1}-\frac{2}{x+2}$ and $\frac{3}{x+1}-\frac{4x}{x^2-1}$.

Working

$$\frac{3}{x+1}-\frac{2}{x+2}=\frac{3(x+2)-2(x+1)}{(x+1)(x+2)}$$

$$=\frac{x+4}{(x+1)(x+2)}.$$

First look for common factors in the denominators to avoid unnecessary multiplication:

$$\frac{3}{x+1}-\frac{4x}{x^2-1}=\frac{3(x-1)-4x}{(x+1)(x-1)}$$

$$=\frac{-x-3}{(x+1)(x-1)}.$$

Test Your Understanding

Solution 23

Edexcel C3 June 2013(R), Q1

Express as a single fraction in its simplest form:

$$\frac{3x+5}{x^2+x-12}-\frac{2}{x-3}.$$

1

$$\frac{3x+5-2(x+4)}{(x+4)(x-3)}$$
$$=\frac{x-3}{(x+4)(x-3)}=\frac{1}{x+4}.$$

Original restrictions: $x \neq -4, 3$.

Express as a single fraction in its simplest form:

$$\frac{10x+4}{3x^2+4x+1}-\frac{3}{x+1}.$$

$$\frac{10x+4-3(3x+1)}{(3x+1)(x+1)}$$
$$=\frac{x+1}{(3x+1)(x+1)}=\frac{1}{3x+1}.$$

Original restrictions: $x \neq -1, -\frac{1}{3}$.

Exercise 1C (Page 8)

Solution 24

Exercise 1C · Pure Mathematics Year 2

Textbook page 8, Chapter 1: Algebraic Methods.

Open Exercise 1C in the textbook

Complete the exercise. Use the techniques for adding and subtracting algebraic fractions from the preceding examples.

Partial Fractions

Solution 25

If the **denominator is a product of a linear terms**, it can be split into the sum of 'partial fractions', where **each denominator is a single linear term**.

$$\frac{6x-2}{(x-3)(x+1)} \equiv \frac{A}{x-3} + \frac{B}{x+1}$$

Notation reminder: $\equiv$ means 'equivalent/identical to', and indicates that both sides are equal for **all** values of x.

Method 1: Substitution

Method 1

$6x - 2 \equiv A(x+1) + B(x-3)$

Let $x = -1$:

$$-8 = -4B \quad \rightarrow \quad B = 2$$

Let $x = 3$:

$$16 = 4A \quad \rightarrow \quad A = 4$$

$$\therefore \frac{6x-2}{(x-3)(x+1)} \equiv \frac{4}{x-3} + \frac{2}{x+1}$$

Multiply both sides by the denominator of the left-hand side.

Choose values of x that make one bracket zero.

Multiplying A/(x−3) by (x−3)(x+1) cancels the factor x−3, leaving A(x+1). The textbook first adds the fractions on the right; either route gives the same identity.

Method 2: Comparing Coefficients

Method 2

$6x - 2 \equiv A(x+1) + B(x-3)$

Comparing coefficients of x terms:

$$6 = A + B$$

Comparing constant terms:

$$-2 = A - 3B$$

Solving simultaneously:

$$A = 4, \quad B = 2$$

$$\therefore \frac{6x-2}{(x-3)(x+1)} \equiv \frac{4}{x-3} + \frac{2}{x+1}$$

If two sides are identical, x terms must match, constant terms must match and so on, e.g. if

$$3x + 4 \equiv ax + b$$

Then $a = 3$ and $b = 4$, whereas this is not necessarily true for a normal equality.

Further Example

Solution 26

Given that

$$\frac{6x^2 + 5x - 2}{x(x-1)(2x+1)} \equiv \frac{A}{x} + \frac{B}{x-1} + \frac{C}{2x+1},$$

find A, B, C.

Working

Clear the denominators:

$$6x^2 + 5x - 2 \equiv A(x-1)(2x+1) + Bx(2x+1) + Cx(x-1).$$

Put $x = 1$: $9 = 3B \Rightarrow B = 3$.

Put $x = -\frac{1}{2}$: $-3 = \frac{3}{4}C \Rightarrow C = -4$.

Put $x = 0$: $-2 = -A \Rightarrow A = 2$.

$$\frac{6x^2 + 5x - 2}{x(x-1)(2x+1)} \equiv \frac{2}{x} + \frac{3}{x-1} - \frac{4}{2x+1}.$$

Test Your Understanding

Solution 27

Edexcel C4 June 2005, Q3(a)

Express $\dfrac{5x+3}{(2x-3)(x+2)}$ in partial fractions. (3 marks)

Working

Write $\dfrac{5x+3}{(2x-3)(x+2)} \equiv \dfrac{A}{2x-3} + \dfrac{B}{x+2}$. Then

$5x+3 \equiv A(x+2) + B(2x-3)$.

Put $x = \frac{3}{2}$: $\frac{21}{2} = \frac{7}{2}A \Rightarrow A = 3$.

Put $x = -2$: $-7 = -7B \Rightarrow B = 1$.

$$\frac{5x+3}{(2x-3)(x+2)} \equiv \frac{3}{2x-3} + \frac{1}{x+2}.$$

Exercise 1D page 8

Solution 28

4. Express $\dfrac{6x+1}{x^2+2x-15} - \dfrac{4}{x-3}$ as a single fraction in its simplest form.

Working

$$\begin{aligned}\frac{6x+1}{x^2+2x-15} - \frac{4}{x-3} &= \frac{6x+1}{(x+5)(x-3)} - \frac{4(x+5)}{(x+5)(x-3)} \\ &= \frac{6x+1-4x-20}{(x+5)(x-3)} \\ &= \frac{2x-19}{(x+5)(x-3)}.\end{aligned}$$

Exercise 1D page 8 — Solution 29

6. Express $\dfrac{4(2x-1)}{36x^2-1}+\dfrac{7}{6x-1}$ as a single fraction in its simplest form.

Working

$$\begin{aligned}\frac{4(2x-1)}{36x^2-1}+\frac{7}{6x-1} &= \frac{4(2x-1)+7(6x+1)}{(6x-1)(6x+1)}\\ &= \frac{8x-4+42x+7}{(6x-1)(6x+1)}\\ &= \frac{50x+3}{(6x-1)(6x+1)}.\end{aligned}$$

Exercise 1D page 8 — Solution 30

7. $g(x)=x+\dfrac{6}{x+2}+\dfrac{36}{x^2-2x-8}$, for $x\in\mathbb{R}$, $x\neq -2, 4$.

(a) Show that $g(x)=\dfrac{x^3-2x^2-2x+12}{(x+2)(x-4)}$.

7a

$$\begin{aligned}g(x) &= x+\frac{6}{x+2}+\frac{36}{(x+2)(x-4)}\\ &= \frac{x(x+2)(x-4)+6(x-4)+36}{(x+2)(x-4)}\\ &= \frac{x^3-2x^2-8x+6x-24+36}{(x+2)(x-4)}\\ &= \frac{x^3-2x^2-2x+12}{(x+2)(x-4)}.\end{aligned}$$

(b) Using algebraic long division, or otherwise, show that $g(x)=\dfrac{x^2-4x+6}{x-4}$.

7b

The numerator is zero at $x=-2$, so $x+2$ is a factor. Division by $x+2$ gives:

$$\begin{array}{r} x^3-2x^2-2x+12 \\ \underline{-(x^3+2x^2)} \\ -4x^2-2x+12 \\ \underline{-(-4x^2-8x)} \\ 6x+12 \\ \underline{-(6x+12)} \\ 0 \end{array}$$

The quotient is x^2-4x+6, hence

$$g(x)=\frac{(x+2)(x^2-4x+6)}{(x+2)(x-4)}=\frac{x^2-4x+6}{x-4}.$$

Repeated linear factors

Solution 31

Suppose we wished to express $\frac{2x+1}{(x+1)^2}$ as $\frac{A}{x+1}+\frac{B}{x+1}$. What's the problem?

Explanation

Because the denominators are the same, we'd get $\frac{A+B}{x+1}$. There's no constant values of A and B we can choose such that $\frac{2x+1}{(x+1)^2} \equiv \frac{A+B}{x+1}$ because the denominators will still be different.

Split $\frac{11x^2+14x+5}{(x+1)^2(2x+1)}$ into partial fractions.

$$\frac{11x^2+14x+5}{(x+1)^2(2x+1)} \equiv \frac{A}{x+1}+\frac{B}{(x+1)^2}+\frac{C}{2x+1}$$

The problem is resolved by having the factor **both squared and non-squared** (explanation of why we do this at the end of these slides).

Working

$$11x^2+14x+5 \equiv A(x+1)(2x+1)+B(2x+1)+C(x+1)^2$$

When $x=-1$: $2=-B \quad \rightarrow \quad B=-2$

When $x=-\frac{1}{2}$: $\frac{3}{4}=\frac{1}{4}C \quad \rightarrow \quad C=3$

At this point we could substitute something else (e.g. $x=1$) but it's easier to equate x^2 terms.

$11=2A+C$

$A=4$

Test Your Understanding

Solution 32

Edexcel C4 June 2011, Q1

Given $\frac{9x^2}{(x-1)^2(2x+1)} \equiv \frac{A}{x-1}+\frac{B}{(x-1)^2}+\frac{C}{2x+1}$, find A, B, C. (4 marks)

Working

$9x^2 \equiv A(x-1)(2x+1)+B(2x+1)+C(x-1)^2.$

Put $x=1$: $9=3B \Rightarrow B=3.$

Put $x=-\frac{1}{2}$: $\frac{9}{4}=\frac{9}{4}C \Rightarrow C=1.$

Compare coefficients of x^2: $9=2A+C \Rightarrow A=4.$

$$\frac{9x^2}{(x-1)^2(2x+1)} \equiv \frac{4}{x-1}+\frac{3}{(x-1)^2}+\frac{1}{2x+1}.$$

Exercise 1E Year 2 page 13 — Solution 33

1. $f(x) = \dfrac{3x^2 + x + 1}{x^2(x+1)}$, $x \neq 0, -1$.

Given $f(x) \equiv \dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{C}{x+1}$, find A, B, C.

Working

$3x^2 + x + 1 \equiv Ax(x+1) + B(x+1) + Cx^2$.

Put $x = 0$: $1 = B$.

Put $x = -1$: $3 - 1 + 1 = C \Rightarrow C = 3$.

Compare coefficients of x^2: $3 = A + C \Rightarrow A = 0$.

$A = 0, \quad B = 1, \quad C = 3; \qquad f(x) = \dfrac{1}{x^2} + \dfrac{3}{x+1}$.

Exercise 1E Year 2 page 13 — Solution 34

4. Show that $\dfrac{5x^2 - 2x - 1}{x^3 - x^2}$ can be written as $\dfrac{C}{x} + \dfrac{D}{x^2} + \dfrac{E}{x-1}$, finding C, D, E.

Working

Factorise $x^3 - x^2 = x^2(x-1)$, then clear denominators:

$5x^2 - 2x - 1 \equiv Cx(x-1) + D(x-1) + Ex^2$.

Put $x = 0$: $-1 = -D \Rightarrow D = 1$.

Put $x = 1$: $2 = E$.

Compare coefficients of x^2: $5 = C + E \Rightarrow C = 3$.

$\dfrac{5x^2 - 2x - 1}{x^3 - x^2} \equiv \dfrac{3}{x} + \dfrac{1}{x^2} + \dfrac{2}{x-1}$.

Dealing with Improper Fractions

Solution 35

In Pure Year 1, we saw that the '**degree**' of a polynomial is the highest power, e.g. a quadratic has degree 2.
An algebraic fraction is **improper** if the degree of the numerator is **at least** the degree of the denominator.

$\frac{x^2-3}{x+2}$ $\frac{x+1}{x-1}$ $\frac{x^3-x^2+3}{x^2-x}$

A partial fraction is still improper if the degree is the same top and bottom.

Questions might take one of two forms:
- Do the division to express as a quotient and a remainder, e.g. $\frac{x+1}{x-1} \rightarrow 1+\frac{2}{x-1}$
- Express as partial fractions, e.g. $\frac{x^2+x}{(x+1)(x-2)} = A+\frac{B}{x+1}+\frac{C}{x-2}$

Reducing to Quotient and Remainder

Solution 36

For example, $7 \div 3 = 2$ remainder 1, so $\frac{7}{3} = 2+\frac{1}{3}$.

In general,

$$\frac{F(x)}{\text{divisor}} = Q(x) + \frac{\text{remainder}}{\text{divisor}},$$

where $Q(x)$ is the quotient.

If

$$\frac{x^2+5x-9}{x+2} = Ax+B+\frac{C}{x+2},$$

determine A, B, C.

Division

Algebraic division

The quotient has degree $2-1=1$, so it has the form $Ax+B$.

$$\begin{array}{r} x^2+5x-9 \\ \underline{-(x^2+2x)} \\ 3x-9 \\ \underline{-(3x+6)} \\ -15 \end{array}$$

Quotient: $x+3$. Remainder: -15.

Result

$$\frac{x^2+5x-9}{x+2} = x+3-\frac{15}{x+2}.$$

$A=1, \quad B=3, \quad C=-15.$

Tip: quotient degree equals numerator degree minus divisor degree. Here $2-1=1$.

Test Your Understanding — Solution 37

Given that

$$\frac{3x^4-2x^3-5x^2-4}{x^2-4} \equiv ax^2+bx+c+\frac{dx+e}{x^2-4}, \quad x \neq \pm 2,$$

find the constants a, b, c, d, e. (4 marks)

Tip: insert $+0x$ for missing powers before dividing.

Division

Algebraic division

Quotient: $3x^2-2x+7$.

$$\begin{array}{r} 3x^4-2x^3-5x^2+0x-4 \\ -(3x^4+0x^3-12x^2) \\ \hline -2x^3+7x^2+0x-4 \\ -(-2x^3+0x^2+8x) \\ \hline 7x^2-8x-4 \\ -(7x^2+0x-28) \\ \hline -8x+24 \end{array}$$

$a=3,\ b=-2,\ c=7,\ d=-8,\ e=24.$

Alternative

Alternative: compare coefficients

Equate the numerator with the divisor multiplied by the quotient, plus the remainder:

$$\begin{aligned} &3x^4-2x^3-5x^2-4 \\ \equiv\ &(ax^2+bx+c)(x^2-4) \\ &+dx+e. \end{aligned}$$

$a=3, \quad b=-2.$

$c-4a=-5 \Rightarrow c=7.$

$d-4b=0 \Rightarrow d=-8.$

$e-4c=-4 \Rightarrow e=24.$

Dealing with Improper Fractions — Solution 38

Split $\frac{3x^2-3x-2}{(x-1)(x-2)}$ into partial fractions.

Method 1: Algebraic Division

Method 1

$$\frac{3x^2-3x-2}{(x-1)(x-2)} \equiv \frac{3x^2-3x-2}{x^2-3x+2}$$

Dividing algebraically gives:

$$3+\frac{6x-8}{x^2-3x+2}$$

Turn numerator back:

$$=3+\frac{6x-8}{(x-1)(x-2)}$$

Let $\frac{6x-8}{(x-1)(x-2)} \equiv \frac{A}{x-1}+\frac{B}{x-2}$

$$A=2$$

$$B=4$$

So $\mathbf{\frac{3x^2-3x-2}{(x-1)(x-2)} \equiv 3+\frac{2}{x-1}+\frac{4}{x-2}}$

Method 2: Using One Identity

Method 2

Let:

$$\frac{3x^2-3x-2}{(x-1)(x-2)} \equiv A+\frac{B}{x-1}+\frac{C}{x-2}$$

$$3x^2-3x-2 \equiv A(x-1)(x-2)+B(x-2)+C(x-1)$$

If $x=2$: $4=C$

If $x=1$: $-2=-B \quad \rightarrow \quad B=2$

Comparing coefficients of x^2:

$3=A$

Test Your Understanding — Solution 39

Edexcel C4 January 2013, Q3

Express $\dfrac{9x^2+20x-10}{(x+2)(3x-1)}$ in partial fractions. (4 marks)

Working

Let the expression be $A+\dfrac{B}{x+2}+\dfrac{C}{3x-1}$. Then

$9x^2+20x-10 \equiv A(x+2)(3x-1)+B(3x-1)+C(x+2)$.

Compare x^2 coefficients: $9=3A \Rightarrow A=3$.

Put $x=-2$: $-14=-7B \Rightarrow B=2$.

Put $x=\frac{1}{3}$: $-\frac{7}{3}=\frac{7}{3}C \Rightarrow C=-1$.

Alternatively, compare x and constant terms: $20=5A+3B+C$, $-10=-2A-B+2C$.

$$\frac{9x^2+20x-10}{(x+2)(3x-1)} \equiv 3+\frac{2}{x+2}-\frac{1}{3x-1}.$$

Exercise 1G (Page 18) — Solution 40

6. Given $\dfrac{2x^2-1}{x^2+2x+1} \equiv A+\dfrac{B}{x+1}+\dfrac{C}{(x+1)^2}$, find A, B, C. (4 marks)

Working

Division gives $2x^2-1=2(x^2+2x+1)-4x-3$. Thus

$$\frac{2x^2-1}{x^2+2x+1}=2+\frac{-4x-3}{(x+1)^2}.$$

Write $-4x-3 \equiv B(x+1)+C$.

Put $x=-1$: $C=1$. Put $x=0$: $-3=B+1 \Rightarrow B=-4$.

$$\frac{2x^2-1}{x^2+2x+1} \equiv 2-\frac{4}{x+1}+\frac{1}{(x+1)^2}.$$

Therefore $A=2,\ B=-4,\ C=1$.

Exercise 1G (Page 18) — Solution 41

7. By factorising the denominator, express each fraction in partial fractions.

(a) $\dfrac{4x^2+17x-11}{x^2+3x-4}$

(b) $\dfrac{x^4-4x^3+9x^2-17x+12}{x^3-4x^2+4x}$

7a

$$\frac{4x^2+17x-11}{x^2+3x-4} = 4+\frac{5x+5}{(x+4)(x-1)}.$$

Write $5x+5 \equiv A(x-1)+B(x+4)$.

At $x=1$: $10=5B \Rightarrow B=2$.

At $x=-4$: $-15=-5A \Rightarrow A=3$.

$$\frac{4x^2+17x-11}{x^2+3x-4} \equiv 4+\frac{3}{x+4}+\frac{2}{x-1}.$$

7b

The denominator is $x(x-2)^2$. Division gives

$$x+\frac{5x^2-17x+12}{x(x-2)^2}.$$

Write

$5x^2-17x+12 \equiv A(x-2)^2+Bx(x-2)+Cx$.

At $x=0$: $12=4A \Rightarrow A=3$.

At $x=2$: $-2=2C \Rightarrow C=-1$.

Compare x^2: $5=A+B \Rightarrow B=2$.

$$x+\frac{3}{x}+\frac{2}{x-2}-\frac{1}{(x-2)^2}.$$

Exercise 1G (Page 18) — Solution 42

8. Given $\dfrac{6x^3-7x^2+3}{3x^2+x-10} \equiv Ax+B+\dfrac{C}{3x-5}+\dfrac{D}{x+2}$, find A, B, C, D. (6 marks)

Working

Divide by $3x^2+x-10=(3x-5)(x+2)$:

$6x^3-7x^2+3=(2x-3)(3x^2+x-10)+(23x-27)$.

Hence $A=2,\ B=-3$. For the remaining fraction,

$23x-27 \equiv C(x+2)+D(3x-5)$.

Put $x=\frac{5}{3}$: $\frac{34}{3}=\frac{11}{3}C \Rightarrow C=\frac{34}{11}$.

Put $x=-2$: $-73=-11D \Rightarrow D=\frac{73}{11}$.

$$\frac{6x^3-7x^2+3}{3x^2+x-10} \equiv 2x-3+\frac{34}{11(3x-5)}+\frac{73}{11(x+2)}.$$

Informal proof of method for repeated factors

If a factor is repeated, why do have a partial fraction with the squared and one without?
When we split into partial fractions, we want each fraction to be non-top-heavy algebraic fractions – recall this means that the 'order' of the numerator has to be less than the order of the denominator. We assume the most generic non-top-heavy fraction possible, i.e. where the order of the numerator is one less than the denominator…

$$\frac{3x^2+1}{(x+1)x^2} = \frac{A}{x+1} + \frac{Bx+C}{x^2}$$

order 1

order 2

Since the denominator is quadratic (order 2), we want the **most generic possible numerator whilst avoiding the fraction being top heavy**, i.e. linear (order 1). Just putting $\frac{B}{x^2}$ wouldn't be good enough because there *might* have been an x term in the numerator.

$$\frac{3x^2+1}{(x+1)x^2} = \frac{A}{x+1} + \frac{Bx}{x^2} + \frac{C}{x^2}$$

Split the fraction.

We can see therefore we have both the x with and without the squared in the denominator.

$$\frac{3x^2+1}{(x+1)x^2} = \frac{A}{x+1} + \frac{B}{x} + \frac{C}{x^2}$$

Informal proof of method for repeated factors

This is easy enough if the repeated factor is x^2, but what about more general repeated linear factors?

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{Bx+C}{(x+1)^2}$$

We want something in the numerator where a $(x+1)$ will cancel, so a bit of clever manipulation is required.

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{B(x+1)-B+C}{(x+1)^2}$$

Split the fraction as before.

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{B(x+1)}{(x+1)^2} + \frac{C-B}{(x+1)^2}$$

$$\frac{3x^2+1}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{D}{(x+1)^2}$$

$C-B$ is just a generic constant, so replace with a single constant.